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**Accreditation by NAAC A+ Grade with CGPA 3.27
(AUTONOMOUS)**

Department of Information Technology

PROJECT IDEA PROPOSAL

MealMentor AI

*An AI-Powered Personalized Nutrition Analysis and Diet Recommendation
Android Application*

Submitted in partial fulfillment of the requirements for the course

Field Project (UIFPR310)

B. Sc. Information Technology — Semester V

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Programme: B. Sc. Information Technology (2026–27) Semester V

Course: Field Project | Code: UIFPR310

Academic Year: 2026 – 2027

CERTIFICATE

This is to certify that the project idea proposal titled “**MealMentor AI: An AI-Powered Personalized Nutrition Analysis and Diet Recommendation Android Application**” has been presented by **Irfan Pathan (SAP ID: 53013240006)**, a student of **B. Sc. Information Technology, Semester V**, at **Usha Pravin Gandhi College of Arts, Science and Commerce (Autonomous), Mumbai**.

The project idea has been reviewed and evaluated for its feasibility, technical soundness, and relevance to the course **Field Project (UIFPR310)** during the academic year **2026–2027**.

The proposed idea is hereby **approved** for further development and implementation under the guidance of the undersigned.

Dr. Prashant Chaudhary

Project Guide

Date: _____

Head of Department

Dept. of Information Technology

Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, **Dr. Prashant Chaudhary**, for his invaluable guidance and constructive feedback during the ideation and proposal presentation of this project. His expertise and encouragement helped refine the concept and scope of MealMentor AI into a well-defined and feasible project idea.

I extend my heartfelt thanks to the **Head of Department** and the **faculty members** of the Department of Information Technology at **Usha Pravin Gandhi College of Arts, Science and Commerce** for providing the platform to present this idea and for their thoughtful suggestions during the review process.

I am also grateful to my family and friends for their unwavering support and motivation throughout this process.

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ABSTRACT

MealMentor AI is a proposed intelligent Android application that leverages artificial intelligence and cloud computing to serve as a personal diet coach. The application will enable users to photograph their meals, automatically identify the food items using computer vision (Azure OpenAI GPT-4o Vision), extract detailed nutritional information, and maintain a comprehensive daily food log.

The core differentiator of MealMentor AI is its **AI Dietitian module** — a context-aware conversational assistant that behaves like a professional dietitian. It will consider the user's complete health profile (age, weight, medical conditions, dietary preferences, allergies), parsed health reports (blood work, cholesterol levels), and the specific foods consumed on that day to provide highly personalized meal recommendations.

The application will feature interactive data visualization through detailed charts — calorie trend lines, macronutrient donut charts, weekly progress bar graphs, and nutrient radar charts. The complete system will be built on **Microsoft Azure cloud services** (App Service, PostgreSQL, Blob Storage, OpenAI Service), ensuring scalability, reliability, and enterprise-grade security.

Keywords: Artificial Intelligence, Nutrition Analysis, Computer Vision, Natural Language Processing, Diet Recommendation, Android Application, Azure Cloud Services, Health Informatics

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CHAPTER 1

INTRODUCTION

1.1 Background

The global burden of diet-related diseases has reached unprecedented levels. According to the World Health Organization (WHO), unhealthy diets are among the top risk factors for non-communicable diseases (NCDs), which collectively account for approximately 71% of all deaths globally. In India, the National Family Health Survey (NFHS-5, 2019–2021) reports that 24% of adults are overweight or obese, while 18.7% of women and 16.2% of men are underweight — highlighting a dual burden of malnutrition.

Traditional methods of food logging — manually searching databases, reading nutritional labels, or maintaining food diaries — are tedious, error-prone, and unsustainable. Research published in the *Journal of Medical Internet Research* indicates that manual food logging has an abandonment rate exceeding 60% within the first two weeks.

Recent advancements in AI, specifically in computer vision and Large Language Models (LLMs), have opened transformative possibilities. Vision-capable AI models such as GPT-4o can now identify food items from photographs with remarkable accuracy, while conversational AI can provide personalized dietary guidance that was previously available only through expensive professional consultations.

MealMentor AI is conceived at this intersection — an intelligent Android application combining AI-powered food recognition, automated nutritional analysis, comprehensive meal tracking, interactive data visualization, and a conversational AI dietitian.

1.2 Motivation

1. **Accessibility Gap:** Professional dietitian consultations cost ₹500–₹2,000 per session in India, making personalized nutrition guidance inaccessible to many. An AI-powered app can democratize this access.
2. **Manual Logging Fatigue:** Existing calorie-tracking apps rely on tedious manual entry. AI image recognition can reduce this friction dramatically.

3. **Lack of Contextual Intelligence:** Most apps provide generic recommendations without considering medical conditions, blood work results, or what the user has already eaten that day.
4. **Cloud Computing Evolution:** Microsoft Azure's ecosystem now offers AI, database, storage, and compute services that make building sophisticated health applications feasible for a college project.
5. **Health Awareness:** The post-COVID era has seen significant increase in health consciousness, creating demand for intelligent tools beyond simple calorie counting.

1.3 Objectives

1. To design and develop an Android application that uses AI-powered computer vision (Azure OpenAI GPT-4o) to identify food items from photographs and extract detailed nutritional information.
2. To implement a comprehensive daily meal tracking system with persistent food logging and progress monitoring against personalized targets.
3. To build an AI Dietitian chat module — a context-aware conversational assistant providing personalized meal recommendations based on health profile, medical conditions, and daily food intake.
4. To create interactive data visualizations (line charts, donut charts, bar graphs, radar charts) presenting nutritional trends over daily, weekly, and monthly periods.
5. To implement health report analysis that allows users to upload medical reports for AI-based parsing, integrating extracted health markers into dietary recommendations.
6. To deploy the backend on Microsoft Azure cloud services, demonstrating proficiency in cloud-native application development.

1.4 Scope of the Project

In Scope:

- Android mobile application using React Native (Expo)
- AI-powered food image recognition and nutritional analysis
- User authentication and health profile management
- Daily meal logging (AI-assisted and manual entry)
- Interactive nutritional charts and analytics dashboard

- Conversational AI Dietitian with context-aware recommendations
- Health report upload and AI-based parsing
- Cloud deployment on Microsoft Azure

Out of Scope:

- iOS application (future extension via React Native cross-platform)
- Wearable device integration
- Barcode/QR code scanning for packaged foods
- Multi-language support (English only in current version)

CHAPTER 2

PROBLEM STATEMENT

The current landscape of dietary management presents several critical challenges that MealMentor AI aims to solve:

No.	Problem	Description	MealMentor AI Solution
P1	Inaccurate Self-Reporting	Studies in the <i>BMJ</i> reveal individuals underreport caloric intake by 30–50% with manual logging methods.	AI-powered food image recognition eliminates manual guesswork; GPT-4o Vision identifies food items and portions from photographs.
P2	High Abandonment Rates	Nutrition apps have an average 30-day retention rate of only 14.3% (Statista, 2024) due to tedious manual food entry.	One-tap meal logging via camera reduces friction; engaging charts and AI conversations maintain user interest.
P3	Generic Recommendations	Most apps provide standardized dietary advice, failing to account for individual health conditions, blood work, allergies, or daily intake context.	Context-aware AI Dietitian considers complete health profile + today's meals + nutritional gaps for hyper-personalized advice.
P4	Disconnected Health Data	Health reports (blood work, checkups) exist in isolation from dietary tracking, with no automated connection to recommendations.	Health report upload feature parses blood work and integrates findings directly into dietary recommendations.
P5	Expensive Professional Guidance	Dietitian consultations cost ₹500–₹2,000+ per session, financially inaccessible for most.	AI Dietitian provides 24/7 personalized nutrition guidance at no per-consultation cost.

CHAPTER 3

LITERATURE SURVEY

3.1 Existing Systems

MyFitnessPal (200M+ users) maintains a database of 14+ million food items with manual entry and barcode scanning, but lacks AI-powered food recognition and contextual intelligence.

HealthifyMe is an India-focused platform offering calorie tracking and human dietitian consultations (premium: ₹999–₹4,999/month). Its “Snap” feature uses basic image recognition but lacks a conversational AI dietitian.

Lose It! offers “Snap It” food recognition from photos but supports limited food categories and lacks context-aware recommendations or health report integration.

Yazio is a European calorie counter with barcode scanning and meal planning but no AI-powered food recognition or conversational dietary guidance.

3.2 Research Papers Reviewed

[1] **Mezgec & Koroušić Seljak (2017)** — “NutriNet” achieved 86.72% top-5 accuracy using CNNs for food recognition across 520 categories. Validates AI-based food recognition viability; MealMentor AI uses more advanced GPT-4o Vision.

[2] **Bossard et al. (2014)** — Introduced the Food-101 dataset (101,000 images, 101 categories), establishing benchmarks for food recognition models.

[3] **Sahoo et al. (2019)** — “FoodAI” proposed multi-task learning for simultaneous food recognition, ingredient detection, and nutritional estimation — validating the pipeline approach used in MealMentor AI.

[4] **Vaswani et al. (2017)** — “Attention Is All You Need” introduced the Transformer architecture, the foundation of GPT-4o and all modern LLMs powering this project.

[5] **Achiam et al. (2023)** — GPT-4 Technical Report details multimodal (vision) capabilities for understanding images alongside text, directly relevant to MealMentor AI’s food analysis.

3.3 Comparative Analysis

Table 3.1: Comparison of Existing Nutrition Applications

Feature	MyFitnessPal	HealthifyMe	Lose It!	Yazio	MealMentor AI
AI Food Image Recognition	✗	Basic	Limited	✗	✓ GPT-4o
Detailed Nutrition Charts	Basic	Basic	Basic	✓	✓ Advanced
Conversational AI Dietitian	✗	✗	✗	✗	✓
Context-Aware Suggestions	✗	Limited	✗	✗	✓
Health Report Integration	✗	✗	✗	✗	✓
Micronutrient Tracking	Premium	Limited	Premium	Premium	✓ Free
Cloud Platform	AWS	AWS	AWS	GCP	Azure

Research Gap: No existing consumer app integrates GPT-4o-level vision models with a conversational AI dietitian that considers daily meal context and parsed health reports. MealMentor AI fills this gap.

CHAPTER 4

PROPOSED SYSTEM

4.1 System Overview

MealMentor AI is a three-tier Android application:

1. **Presentation Tier:** React Native (Expo) mobile app with 10+ screens.
2. **Application Tier:** Node.js + Express RESTful API deployed on Azure App Service with 15+ endpoints.
3. **Data Tier:** Azure PostgreSQL (pgvector), Azure Blob Storage, and Azure OpenAI Service (GPT-4o + ada-002).

4.2 Key Features

#	Feature	Description
1	AI Food Recognition	Photograph meals → GPT-4o Vision identifies food items, estimates portions, extracts calories, macros (protein, carbs, fat), and micronutrients (vitamins, minerals).
2	Daily Meal Tracking	Every meal logged with timestamp, meal type, nutritional data, and photo. Dashboard shows real-time progress toward daily targets with circular progress indicators.
3	Interactive Charts	Calorie trend line chart, macronutrient donut chart, weekly stacked bar chart, and nutrient radar chart — with daily/weekly/monthly views.
4	AI Dietitian Chat	Conversational AI with access to user's complete health profile, today's meals, nutritional gaps, and health reports. Provides personalized meal advice.
5	Health Report Parsing	Upload blood work/medical reports → AI extracts cholesterol, blood sugar, vitamin levels, thyroid markers → factors into all dietary recommendations.
6	Proactive Suggestions	Based on time of day and nutritional gaps, app suggests 3 meals that optimally fill remaining nutritional needs while respecting all health constraints.
7	Meal History	Searchable calendar view of all past meals with complete nutritional data for long-term tracking.

4.3 AI Dietitian — Core Differentiator

The AI Dietitian is the flagship feature that sets MealMentor AI apart. Every AI interaction builds a rich context from:

1. User profile (age, weight, conditions, preferences, allergies)
2. Parsed health report data (blood markers, vitamin levels)
3. Today's complete meal log with nutritional totals
4. Calculated remaining nutritional budget (calories, protein, carbs, fat)
5. Identified nutritional gaps (nutrients below 50% daily value)
6. 7-day meal history for pattern recognition

Example Interaction:

"I had poha for breakfast and chole chawal for lunch. What should I eat for dinner?"

→ The AI calculates remaining caloric budget (~600 cal), identifies low protein (35g of 80g target), and suggests high-protein dinners respecting the user's vegetarian preference and lactose intolerance, with specific nutritional breakdowns for each suggestion.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 High-Level Architecture

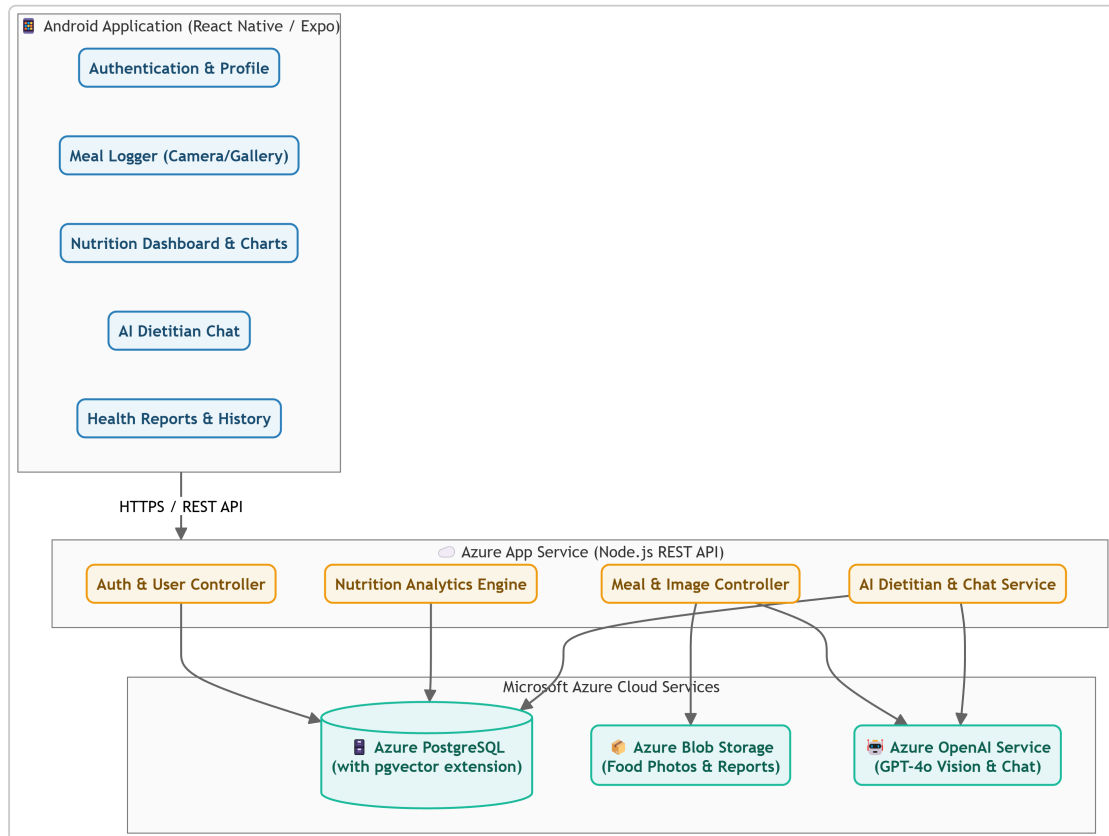


Figure 5.1: High-Level System Architecture of MealMentor AI

5.2 Key Data Flows

Food Image Analysis Flow

```
User captures photo → App sends to API → API uploads to Blob Storage
→ API sends image to GPT-4o Vision → GPT-4o identifies food + nutrition
→ API stores in PostgreSQL (meals + nutrition_entries tables)
→ API generates personalized recommendation using user profile context
→ Results returned to app with charts-ready data
```

AI Dietitian Chat Flow

User types message → App sends to API → API fetches from DB:

- User profile + health conditions + allergies
- Today's meals + nutrition totals + remaining budget
- Health report markers + recent chat history

→ API builds context-rich system prompt

→ Sends to GPT-4o → Response returned to app → Stored in chat history

5.3 Database Overview

The database consists of **6 tables** built on PostgreSQL with the pgvector extension:

Table	Purpose	Key Columns
users	User profiles & health data	email, age, height, weight, goals, dietary_preferences, allergies, health_conditions, embedding
meals	Logged meals	user_id, meal_type, name, image_url, logged_at
nutrition_entries	Per-item nutrition data	meal_id, food_name, calories, protein, carbs, fat, vitamins, minerals
health_reports	Uploaded medical reports	user_id, report_type, file_url, parsed_data (JSONB)
chat_messages	AI Dietitian conversation	user_id, role, content, created_at
daily_summaries	Pre-computed daily totals	user_id, date, total_calories, total_protein, total_carbs, total_fat

CHAPTER 6

TECHNOLOGY STACK

Table 6.1: Complete Technology Stack

Layer	Technology	Purpose
Frontend (Mobile App)	React Native + Expo	Cross-platform Android app framework
	React Navigation 6.x	Screen routing & navigation
	Victory Native / Chart Kit	Interactive charts & data visualization
	React Native Gifted Chat	Chat UI for AI Dietitian
	Expo Camera + Image Picker	Camera access & gallery selection
Backend (Server)	Node.js 20 LTS	JavaScript runtime
	Express.js 4.x	REST API framework
	bcryptjs + jsonwebtoken	Password hashing & JWT authentication
	multer	File upload middleware
	pg (node-postgres)	PostgreSQL client
Azure Cloud Services	Azure App Service (B1)	Host Node.js backend API
	Azure Database for PostgreSQL	Primary database with pgvector extension
	Azure Blob Storage	Store food images & health reports
	Azure OpenAI Service	GPT-4o (vision + chat) & ada-002 (embeddings)
AI / ML	GPT-4o	Food image analysis, dietitian chat, report parsing
	text-embedding-ada-002	1536-dim vector embeddings for similarity matching

6.1 Hardware & Software Requirements

Component	Development	End User
OS	Windows 10/11, macOS, or Ubuntu	Android 10+ (API 29)
Processor	Intel i5 / Ryzen 5 or above	Any modern mobile processor
RAM	8 GB minimum	4 GB minimum
Storage	50 GB free space	200 MB for app
Network	Broadband internet	4G / Wi-Fi
Camera	N/A	8 MP or higher
Tools	Node.js 20, Android Studio, Expo CLI, VS Code, Git	Expo Go / APK

CHAPTER 7

IMPLEMENTATION PLAN & TIMELINE

Table 7.1: Phased Implementation Schedule (20 Weeks)

Phase	Duration	Key Tasks	Deliverables
1. Setup	Week 1–2	Azure services provisioning; project initialization; database schema	Working dev environment
2. Auth & Profile	Week 3–4	User registration/login (JWT); profile setup screen & API	Functional auth flow
3. Food Analysis	Week 5–7	Camera UI; GPT-4o Vision integration; nutrition parsing; meal storage	Photo → nutrition pipeline
4. Tracking	Week 8–9	Dashboard with progress indicators; daily summaries; meal history	Complete meal tracking
5. Charts	Week 10–11	Line chart, donut chart, bar chart, radar chart; trends API	Analytics dashboard
6. AI Dietitian	Week 12–14	Chat UI; context builder; GPT-4o integration; meal suggestions	Working AI dietitian
7. Health Reports	Week 15–16	Report upload; AI parsing; health marker storage & integration	Health report feature
8. Polish	Week 17–18	Animations; error handling; testing; performance optimization	Production-quality app
9. Deploy	Week 19–20	Azure deployment; signed APK; project documentation; demo video	Final deliverables

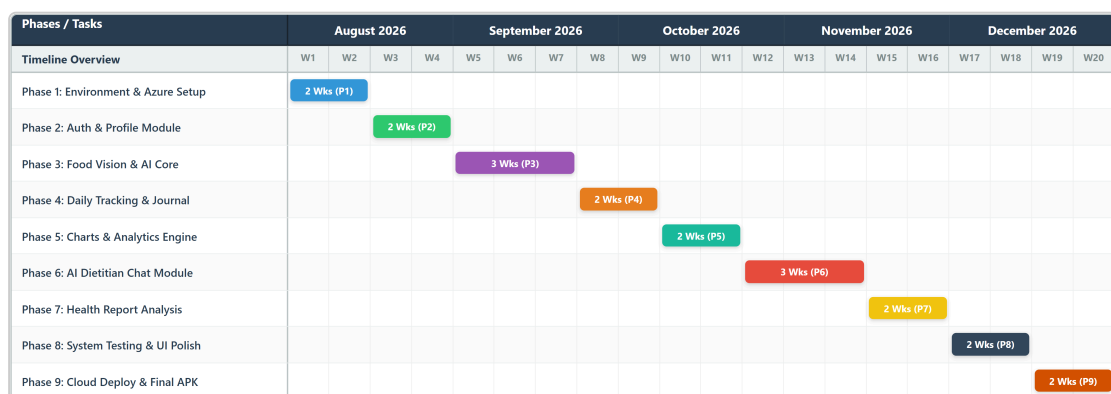


Figure 7.1: Gantt Chart — Project Implementation Timeline

CHAPTER 8

EXPECTED OUTCOMES & FUTURE SCOPE

8.1 Expected Outcomes

1. A **fully functional Android application (APK)** enabling users to photograph meals and receive instant AI-powered nutritional analysis with at least 85% food identification accuracy.
2. An operational **AI Dietitian** providing personalized, context-aware dietary recommendations that simulate professional dietitian consultation quality.
3. **Interactive nutritional dashboards** with four chart types (line, donut, bar, radar) transforming raw data into actionable visual insights.
4. A **scalable cloud-native backend** deployed on Microsoft Azure, demonstrating proficiency in modern cloud computing practices.
5. **Health report parsing** that bridges the gap between clinical data and daily dietary choices.

8.2 Future Scope

1. **iOS Application:** Leverage React Native's cross-platform capability to publish on Apple App Store.
2. **Barcode & Label Scanning:** Integrate barcode scanning for packaged foods via Open Food Facts API.
3. **Wearable Integration:** Connect with Google Fit/Samsung Health for activity-aware dietary recommendations.
4. **Weekly Meal Plans:** Generate complete meal plans with grocery lists and recipes.
5. **Multi-Language Support:** Hindi, Marathi, Tamil, and other regional languages.
6. **Restaurant Menu Integration:** Analyze Zomato/Swiggy menus to suggest healthiest options.
7. **Predictive Health Alerts:** Use dietary trends to predict and prevent health risks.
8. **Offline AI Mode:** On-device inference using TensorFlow Lite for basic recognition without internet.

CHAPTER 9

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